Wyatt Corcoran.
AI Analysis:

1. GenAI Names: Gemini by Google and ChatGPT by OpenAI.
2. Both accessed via Google search.
3. Prompt:
   -Start by ignoring all comments in the input and output; for the following instructions, return code blocks, not a full rewrite, to implement the following change to the program's file I/O functionality:
   Write the solutions for each puzzle file to a single file named solutions.txt.
   - For each puzzle, the solutions.txt file should include the following:
   - A single line: “Puzzle: x”, where x is the puzzle file name (e.g., puzzle1.txt).
   - If no solutions are found, a single line: “No solution found.”
   - If solutions are found, it should include all solutions followed by a single line: “Solutions found: n”, where n is the number of solutions found.
   - The solutions should be in the same format as the input files: puzzle1.txt – puzzle5.txt.
   Other notes: the current iteration of the program outputs each answer to its own respective file; this should be one single file including only the aforementioned formatting. Attempt to make the changes as simple and minimally invasive as possible. Add no includes unless mandatory.
4. Generated code included in respective .txts
5. For this round, I was very specific in making the changes as minimally invasive as possible and included some extra requests to exclude as many extra coding artifacts that gen AI tends to leave in code. Due to the extreme specificity of the request, I was provided with near-identical output from both LLMS. Both LLMS also provided proper block structure as requested. The changes they make to Main are completely identical; it is a small rewrite from the original, changing the original iteration method from opening and closing a file each time to instead writing in the same file. They both also made the same change to the solver helper function, adding the additional formatting text to the file. After some combing, the key difference is that the code from ChatGPT adds an extra new line, which makes the formatting inconsistent with the previous versions of the code.
6. I chose to implement Gemini’s code based on correctness; the output from Gemini’s code more accurately fits the assignment’s output requirements. In terms of maintainability, Gemini edges out ChatGPT slightly since the code provided is better formatted and more readable. Gemini also included more reasoning in the complete output, making implementation possible even without providing the code as the explanation was simple and easy to parse. It is a tie between the 2 for both complexity types since they both use identical methods in their respective outputs. The output is now in a far simpler format; this, by nature, makes the code more maintainable, as it makes checking test cases a simpler process.
7. Implementation successful. Changes include the following: changing the solve member function and the main function to alter the existing file I/O format to output to a single file

with the correct headers to show which puzzle, solution num etc is displayed. The following are aspects of the code left unchanged from the previous iteration; all of the recursion logic from the original code is the same. The previous editor added some error handling for the isvalid function to make the code run better and fix a few hypothetical edge cases. All of the core logical code was unchanged, as it functions very efficiently. The File I/O system from the previous iteration functions perfectly and was unchanged; only the formatting of the output of the file i/o was changed to account for the new output type. No new includes were added. Due to the new case handling, the space and time complexity are as efficient as possible for this particular algorithm.

8. Compiling and debugging were successful; some of the particulars are included above.